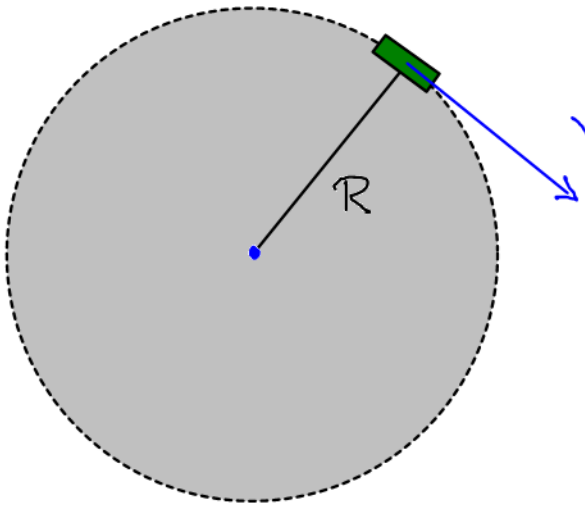
Lösung: Zentripetalbeschleunigung in Kurve



a) $\vec{a}_{zp} = \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad \Rightarrow \quad a_{zp} = \omega^2 \cdot R$

mit $\omega = v/R$ folgt $\boxed{a_{zp} = \frac{v^2}{R}}$

$$a_{zp} \approx \frac{(20\ m/s)^2}{100\ m} \approx 4\ m/s^2$$

b) $$\frac{a_{zp}}{g} \approx \frac{4\ m/s^2}{9{,}81\ m/s^2} \approx 0{,}408 \approx \underline{\underline{40{,}8\ \%}}$$